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Folkmann, Mads Nygaard; Graesse, Malin

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Liquid Forms: Questions for a Design History of Water

Mads Nygaard Folkmann and Malin Graesse

Abstract

The article introduces a largely overlooked topic within the history of design: Designed objects serving as interfaces with water, enabling interaction, managing and control. By investigating how design objects as human-made, 'cultural' artefacts negotiate the meaning of the 'natural' substance of water, we position our thinking within what has been known as 'naturecultures' but expand the discussion by drawing on a theoretical framework of design aesthetics.

The article will first analyse the ontological foundation for investigating design and water in relation to each other. Subsequently, we present our findings as three questions, which together compile a conceptual framework for a design history of water: 1) By historically and phenomenologically contesting how design evokes and affords sensory engagement with water, we can understand design's contribution to humans' relationship with water. 2) By examining the material composition of designed objects of carrying, distributing and accessing water, we can trace the historical change in the materials used. 3) The question of temporal flow facilitates an investigation of how designed means are used to handle, organise and present the fluidity of water.

Keywords

aesthetics – grey ecology – product design – twentieth century – urban design – water

Introduction

As part of antiquity's theory of four basic elements – water, earth, fire and sky – which was effective in European cultural representations until the late 18th century,[1] water had mythological agency and was met with awe.[2] With the scientific revolutions of the late 18th century and onwards, however, water became an object of scientific abstraction to be cultivated, domesticated and disciplined.[3] In design history, the Modernist movement of the 20th century, which sought to reform (Western) societies by means of design and architecture, an ideological peak of managing water for human needs was reached. With improvement of hygiene as a guideline, devices for containing and making water easily accessible were developed. Paradigmatic examples are Le Corbusier's 1929 *Villa Savoye* with designs for running water for common use, such as a sink for washing hands in the entrance hall, and functionalist conceptions of rational and hygienic kitchens, such as in the 1923 Bauhaus exhibition house *Haus am Horn* and the modular 1926 *Frankfurter Küche*.[4]

As such, 'taming' the 'wild' element of water has improved living and health conditions for billions of people; yet, much needs to be done as many people still do not have access to clean water. At the same time, the present critical situation of often having either too much or too little water calls for renegotiating our relationship with it and, in the future, challenge the ways design is used to interact with and understand water. However, while design and designers are becoming increasingly essential to the planning and handling of water – evidenced in the growing number of water-themed courses taught at design and architectural schools globally – the relationship between water and design remains quite understudied in the humanities. Until now, little work has been done on the many ways water is accessed, managed and experienced through design.[5]

In this article, we will attempt to outline how design history and design studies can deal with this complex and dynamic relationship between water and design. By thinking within the framework of so-called 'naturecultures'[6], we will expand the discussion on the perceived dichotomy between natural and artificial by drawing on a theoretical framework of design aesthetics. This means that by understanding design solutions for handling water as interfaces between humans and water, we can explore what happens when design and water meet in an interchange of modalities. Because the aesthetic dimension of design expands discussions on ontological categorisations, which we unfold in the first section, the article will argue that the ontological negotiation between water and design takes place on three different levels: sensory, material and temporal.

Consequently, the second section presents our initial findings for a design history of water in the form of three questions. First, by historically and phenomenologically contesting how means of design evoke and afford *sensory engagement* with water, we can understand design's contribution to humans' relationship with water. Second, by examining the *material constitution* of designed means for carrying, distributing and accessing water, we not only grasp the historical changes in materials employed but also address ontological questions about how water and supporting materials affect each other. Third, the question of *temporal flow* facilitates an investigation of how designed means handle, organize and present the fluidity of water. In combination, these three questions compile a conceptual framework for a design history of water. We will showcase the richness of water-related design by providing a variety of paradigmatic examples for each question.

Natural and Artificial Ontologies

To start with the question of ontology, water is considered 'natural' and 'fluid', and design as 'artificial' and 'solid'. However, these ontologies of the natural and artificial unfold in a dynamic and perpetual interchange with each other. As such, the design/water relationship taps into the core of today's discussions about naturecultures. Our approach is to question the

dichotomous separation of nature and culture and investigate how things like the material and immaterial, process and matter, and biology and nurture are interconnected and work together. We will start by examining the ontologies of water and design individually, and then discuss how they relate.

Separating Ontologies

In its constitution, *water* is highly transitory and appears in many forms. In its liquid state, water is not only impossible to fixate and can find its way through even the tiniest crack, but water molecules are also in constant motion, which creates friction and, in turn, energy. Water, then, always tends to move from suspension towards a state of high energy where the liquid gas evaporates. For this reason, water can go from a highly formable state, such as ice, exemplified by the ice hotel in Jukkasjärvi, Sweden (Fig. 1), to a completely effervescent state. While ice is highly malleable and has a long history of being used and commodified for various purposes, ranging from building materials to cooling mechanisms, water in its gaseous state is less formable but indirectly present in products such as thermometers and hygrometers (Fig. 2), showcasing how a fine-tuned exchange between technology and environmental factors are at work in the designed interfaces between people and water in all its forms.

At the same time, as pointed out by geographer Jamie Linton, what water 'is' or seems to be is largely framed by our notions of it. Under the label 'modern water', Western thought has since the 18th century developed a way of 'knowing water' by abstracting and reducing it to the chemical formula H_2O.[7] In this process of abstraction, which has become habitual, water becomes universal, natural and 'not complicated by ecological, cultural, or societal factors'[8]. In Linton's analysis, the idea of modern water has fixated the abstraction of water in scientific discourse and, furthermore, its 'conceptual abstraction and technical control' has caused a process of deterritorialisation and 'placelessness':[9] When water comes out of a tap, we do not know where it comes from. As such, the modern, abstract *representation* of water (as 'H_2O') in the 'modern idea of water as an objective, homogeneous, ahistorical entity devoid of cultural content'[10] is wrongly being equated with its *essence*. Linton's ambition is to stop taking water for granted as a definite 'thing'. By seeing water as a process, he aims to cast light on the 'ideas, meanings, values, and potentials that we have conferred upon it.'[11]

In contrast, design can be understood as a product[12] and technique of culture,[13] and as artificial creations with an origin in human *intention*. Understanding design as partaking in a Modernist progressive-optimistic ideology, Herbert Simon claimed in 1969 that the 'world we live in today is much more a man-made, or artificial, world than it is a natural world'[14]. Even more to the point, Otl Aicher, co-founder of the Hochschule für Gestaltung Ulm, proposed that design is 'producing the world.'[15] Even if a central motivation for this conception of

design was a desire to envision new possibilities for humanity against the backdrop of Nazi barbarism, it comes with a neglect of non-human otherness. In his most radical formulations, the result of design is, to Aicher, a 'human-made and organized world' which nature becomes part of and must 'submit' to.[16] Similarly, critic Clive Dilnot has suggested that in 'the Anthropocene', 'the artificial, and no longer nature, constitutes the horizon, medium and determining condition of the world.'[17]

Conversely, the notion of human intentionality in design is severely challenged by approaches in theory and practice to integrate living organisms as materials for design,[18] thus blurring the boundaries between the 'natural' and the 'artificial'. A model for contesting the agencies of the 'material' (water) to be handled in a 'form' (design) can be found in the work of the French phenomenologist Gaston Bachelard. In his 1942 book *L'eau et les Rêves*, he explored how water affects literary imagination as part of a series focusing on the cultural impact of antiquity's four elements.[19] In his analysis, Bachelard investigates water as a motif and points out how the sound of water may be reflected in poetic language. More radically, Bachelard posits the flow of water beside the flow of language and 'when the language of water is a direct poetic reality'[20] to affect poetic language to be in flow and dissolution. Of course, poetic texts have a different ontological constitution than material objects of design. But if we, like Bachelard, regard 'the material imagination of water to be a specific kind of imagination'[21], we can, nevertheless, pose the question of how water affects the constitution of design.

Similarly, the advent of new technologies has affected modes of expression of design. One example is how electricity led to innovations of design in the context of Peter Behrens' work for AEG from the beginning of the 20th century, where electricity not only provided new possibilities for designing for new functions and, hence, new types of products – it was also an 'invisible' source of energy that called for being expressed in entirely new ways.[22] Likewise, digital technology has evoked new kinds of objects *and* fostered a need to be represented in the objects in new ways.[23] In this vein, water is not just an abstract substance to be managed in pipes and irrigation systems and to function as a silent servant in households and cities – or, to put it briefly: to be contained by design. The question remains what the processual agency of water does and means for the ways it is designed.

Negotiating Ontologies

The meeting of the 'natural' element of water and the 'artificial' means of design by no means implies a clear division of design as an ideally conceived 'form' for distributing and facilitating the 'substance' of water, but it is a messy business: Water will always find its own way out, waterpipes crack, water causes wear and tear on most materials, and water is rarely in a state of purity. In its present imbalances of too much or too little, water may render

human intentions of any kind ultimately powerless (just think of the 2004 Indian Ocean tsunami).

Conceptualised as an ecology, which refers to the *logia* of *oikos* (the study of household systems), a concept of *grey ecology* is well-suited to understanding how the relationship between culture and nature is negotiated in relation to water. As proposed by the philosopher Paul Virilio, we are not only dealing with 'green' ecology as a theory related to 'pollution of substance',[24] which eventually builds on (utopian) ideals of the sustainable circulation of pure substances. Instead, grey ecology signifies modalities of exchange, blurring the boundaries of what is being exchanged. Explicated by literary scholar Jeffrey Cohen, grey ecology obscures contrast and copular logics. By being an 'open aesthetic' dealing with 'inhuman' and 'non-human'[25], it is transformative in effect. As Cohen argues, a 'green ecology judges a culture by its regard for nature, where "nature" is typically regarded as an external entity, culture's other. A grey ecology refuses such separations'[26].

Cohen's conception of an ecology operating in greyness is informative in two instances. First, the concept of grey ecology can be seen as an attempt to deconstruct human perspectivism in ecological thinking. Grey flattens human perception and turns materials, surfaces and spatial qualities into absorbed appearances. Designating a lack of colour and contrast, whereby boundaries and categorisations become less obvious, grey ecology offers a way of describing the world where humans just become one of many players in the game, and where many modes of existence may participate at the intersection of life and death.

Second, the concept of grey ecology is a way of sidestepping normative judgements of 'nature' or ecological processes, where disasters and negative consequences of environmental entanglements are mostly disregarded or left understudied. In contrast, the greenness of green ecology is not only an expression of human perception and interpretation (the chlorophyll of the plant does not absorb the green light, which is then transmitted to our retinas as green), but also embodies an idealism of nature's purity. This evokes all sorts of ideas of what 'nature' should be in relation to humans.

In this context, we will employ the concept of grey ecology to frame how design solutions related to water participate in the negotiation of (so-called) nature and (so-called) culture where established dichotomies may be set in motion. Further, as pointed out by philosopher Peter Sloterdijk in his 'colour theory' of grey, grey is associated with the inconspicuous, the amorph, the foggy, and the monotone, to all the 'Mitteldinge'[27] of the world, which has been overlooked by traditional aesthetic theory in its favouring of the special and conspicuous. Hereby, we may gain focus on processes of exchange related to water taking place in and with all sorts of everyday objects and solutions ranging from glasses to grids, from visible objects of interaction to hidden infrastructures. The muddiness of the grey will take into consideration the intersection between the present and the absent; between

material and conceptual; and between human and non-human agencies in relation to design and water.

Aesthetic Modalities of Design for Water

Based on the ontological intricacies of design and water, the next step is to investigate the aesthetic modalities of designed objects operating as interfaces between humans and water. Aesthetics as an approach to analysing and conceptualising design reflects, in its essence, an interest in the means and effects of design in relation to human experience. Originally named as a philosophical discipline for sensory experience in the eighteenth century by Alexander Baumgarten, aesthetics has since entered a variety of discourses and manifestations. In this context, Design Aesthetics can comprehend the ability of design objects to frame human experience by the ways sensual appearances (form, material, colour, etc.) relate to conceptual meaning (e.g. what objects are for and the idea).

In brief, Design Aesthetics offers a framework for analysing sensual, conceptual and contextual dimensions of the aesthetics of the objects in question: Not only a positive, sensually appealing appearance of the objects, which traditionally has been associated with the aesthetics of design,[28] but also the conceptual dimension of the staging of meaning by design objects and their ability to evoke a reflection about them (and not just how form relates to function)[29]. Additionally, the contextual dimension embeds aesthetics in a consideration of cultural and societal conditions. Anchored in a phenomenological reflection, the aesthetic question in design deals with the role of objects for human experience.

By focusing on appearances of human-made design, Design Aesthetics allows for criticising the artificial ontology of design. Accordingly, design is not seen as a kind of super-tool for world-making but comprehends the constituents of what makes 'the world' in specific contexts and at specific times. Thus, through this lens, design can be seen as a dynamic and potential artificiality of how 'the world' (in its human-made appearances) is made present for us. In the words of philosopher Martin Seel, aesthetics lets us grasp how artifacts may produce not only 'ways of *meeting the world*' but also 'ways of *meeting the meeting of the world*'[30]. For Seel, this kind of meta-experience – that is, an experience about what 'experience' is – also offers an opportunity of meeting some 'otherwise unrepresentable circumstances'[31]. In Seel's view, this kind of aesthetic experience casts us back onto our own experiences and to the boundaries of those experiences, to their Other.

In relation to water, design objects offer various 'artificial' means to make the 'natural' substance of water present for us and to facilitate our ways of meeting it and, possibly, also meeting our meeting with it. Whether it is city planning, irrigation systems, the commodification of drinking water, wastewater management or social and environmental aid, design plays a crucial role. Furthermore, design gives form and function to everyday objects

related to water. In this instance, design is a central interface between humans and water, especially freshwater. From this perspective, design is as much a cultural activity of meaning attribution as it is a functional means of providing access to water.

Taking the concept of grey ecology into consideration, next, we may pose the question whether (or how) water is to be regarded as a natural 'otherness' being handled or attempted disciplined by various modalities of design objects. Based on this, we will propose three questions for a history of design's mediation of (the otherness of) water: 1) What is design's changing affordances of *sensory engagement*? 2) How has the *material constitution* of design changed? 3) Which different approaches to staging of the *temporal flow* of water by means of design can be found? We will present the theoretical foundations for these questions in turn.

Sensory Engagement

With the ambition of managing water and bringing it into and out of households, the process of modernisation in the 20th century made water more easily accessible in, for instance, kitchens and bathrooms. At the same time, in a reverse movement, water was hidden away in the infrastructures of pipes leading to the access points. As an example, bathrooms can be seen as perforated system of water circulation, oscillating between making water invisible/intangible and visible/tangible. The non-sensory presence of water can be seen in relation to what Linton calls the 'placelessness of modern water', which also transfers 'water control to placeless discourses of hydrological engineering, infrastructural management, and economics'[32].

In contrast, design objects like glasses, bottles, jugs, measuring cups, bowls, buckets, and other containers function as devices for concretely handling water (or other liquids). In this process, they make water visible and accessible for sensual interaction in various ways. In bringing water to appearance, much in the same way Vilém Flusser claimed that bottles bring the specific character of 'beverages to expression',[33] such containers 'place' water. In addition, when being contained, water may itself contribute to the being and effect of its container. In a central passage in his late work *L'Œil et l'Esprit*, phenomenologist Maurice Merleau-Ponty points out how the water in a pool is constitutive for visually perceiving the geometrical structure at the bottom of the pool: How water, 'this syrupy and shimmering element', is not only 'in the space' of the pool, but 'inhabiting it, materializing itself in it, not contained in it' and, next, extends this effect beyond the pool to the 'network of reflections' playing on the surrounding trees, revealing its 'active and living essence'[34]. Of course, a drinking glass may be empty (and so may the pool), but what Merleau-Ponty's reflection reveals is the fundamental co-constitution of the ontologies of container and contained. Furthermore, it highlights the embeddedness of container and contained in a wider setting,

which may also allow visual perception synaesthetically to expand into a kind of tactile feeling of the material.

By taking a closer look at two drinking glasses, we can approach the contribution of design in the sensory engagement with water (Fig. 3). As glassware, a common feature of the two products is their ability to offer an effect of transparency, allowing a clear view of the contained liquids. Furthermore, the specific effects of transparency are produced by the colour of the glasses in combination with the structure of the surface. In this way, a key feature of the industrially manufactured glass from the Finnish company Iittala is the ribbed structure, which creates a regularity of the fractures when viewing the liquids through the glass, much in the same way as when a drop of water makes regular ripples in a water surface. The glass is also available in a variety of colours (including a clear version), and even in the monochromatic glasses, the ribs produce an effect of colour grading, depending on the thickness of the glass along the surface. In contrast, the irregularly shaped, mouth-blown glass is an experimental, uniquely crafted product. It offers not only micro-sensations of touching all the folds and variations of the material surface, but also an ever-changing perception of the movements of the contained liquid. This glass is based on a dogma of using only leftover glass from the production, which makes the colour unpredictable, albeit within a range of blue and green colours, because only these can endure an oven temperature of 1,100 degrees Celsius when preparing the glass for blowing.

Naturally, these two examples have in common that they stand out in comparison to more neutral glasses. They may garner attention and thus evoke a conceptual reflection on their ability to be forms containing a substance (water/liquid), which, in turn, attracts attention to this substance. At the same time, part of the visual effect of the glasses is to reflect the light, making them appear as meeting or transmission points between light and water.

Working with colours and surfaces is a central aspect of the design history of producing glass. We will highlight further examples in two directions. With regard to the work with *colours*, mass-manufacture of monochrome glass reached a turning point and, accordingly, new possibilities of expression through the invention of super-solid, tempered glass by the French company Duralex in the 1960s, enabling design of sets of glass, cups, and plates in coloured glass. In trendy colours of its age, such as amber, smoky grey and sky blue, these products create an encapsulating frame for a complete experience of colouring food and beverages.

With regard to the *structure of the surface*, glasses may, in addition to the tactile sensation of holding the glass, create fractured meeting points of the appearance of liquid and light in various ways. An example of this is Iittala's 1960s Ultima Thule series of pitchers and glasses. The surface of the products appears like abstract mimicries of frozen and liquid water. On the aesthetic level, the product performs a *transformation* of the endlessly

malleable element of water into solid forms of glass, which itself is malleable during the manufacturing process and continues to move ever so slowly, even after fixation. In such products, expressions of nature are interpreted and incorporated into expressions of culture, whereby the agency of water further enters the ontology of the design object.

In these briefly described cases, the drinking glasses *do something* to the liquid they contain – or, to be more precise: their design enhances the aesthetic experience of water by engaging with (the means of engaging with) it sensorily. Such processes do not, however, take place in a cultural and societal vacuum. As a cultural process of making (or conceiving something as) aesthetic in appealing to sensory engagement, *aestheticization* is always at play. As examples, we will point to two different effects of aestheticization.

First, aestheticization deals with creating aesthetic appeal through the surfaces of the objects. Being closely affiliated with commercialisation and consumption, this effect has been criticised as the seducing powers of 'commodity aesthetics'.[35] In our view, such an effect is very much present in commodified water and is especially detectable in highly profiled water. An example of this is the Norwegian company VOSS (Fig. 4), which promises 'to evolve and bring new hydration experiences to the market'. As in (almost) every product building on branding and outer appearance, the claim is also that it reflects quality; the product is not only one of 'outer beauty' but also has 'beauty on the inside including naturally pure, crisp, refreshing still water'[36]. Ultimately, however, this kind of product is more about branding than the liquid content. It is, thus, an example of the tendency in late-modern societies to foster highly profiled products because consumers demand the 'singularity' of special experiences.[37] In the end, this kind of surface-oriented and, hence, 'superficial' aesthetics is intimately connected to driving the desire for consumption.

According to the philosopher Wolfgang Welsch, who has proposed a theory of aestheticization at the intersection of cultural theory and philosophy, this focus on seductive appearances is typical for 'surface aestheticization' working with 'sugarcoating of reality' through beautification and immediate sensual appeal.[38] In contrast, he proposes a concept of 'deep aestheticization'. It builds on the assumption that not only things, forms and surfaces are being aestheticized but also the perception of the reality and the 'structure of our understanding of reality [preceding] all form-giving [Gestaltung]'[39]. In this way, the effect of aestheticization extends beyond surfaces, creating new conditions for experiencing the meaningful properties of the matter in question. When water is matter and the drinking glass is the vehicle for aestheticizing it, the result can be a new, sensual and conceptually rich experience of the meeting with (the meeting of) water.

Of course, using drinking glasses is a part of everyday, habitual experience and rarely considered something special. Our point is not, that we need to seek and elevate ordinary experiences of drinking water to extraordinary ones;[40] nor do we need to remark on every bit

of the process. Doing so would result in sensual overload. Instead, we see a potential in recognising and historically investigating the role of various design means for creating sensory engagement with water, for instance through transparent, coloured and fractured glass, but without necessarily noticing these as 'designed' means.

Materiality

Materially, the relationship between design and water includes and relies upon a myriad of (in lack of a better word) 'support' materials. Clearly, it is the glass, as discussed above. However, it is also the concrete and the plastic that contain, distribute and manage water, which, from a design perspective, make up the material interface between us and water. It is the concrete of costal shielding and the plastic in a water bottle that make up the designed object. We may, of course, experience water by having it splashed over us or by swimming in the sea, but in many instances our interaction with it is mediated through a material interface.

While water always is in motion and 'in process'[41], the materials used to contain, facilitate, hinder, control (or attempt to control) and steer water are fixed in character. These materials can be solid, like metals in taps and concrete in fishways, or malleable, like soft plastics or rubber used in pipes. In terms of materials ranging from solid to soft, we may find various conceptually tinted designs, like Rob Brandt's 1975 *Crinkle Cup* made of the hard material ceramic but imitating a soft plastic cup. In the end, materials used in relation to water may be solid or flexible, but they contrast with the flow of slippery water and create barriers for it, unlike most textiles.

The role of materialities has recently gained new interest. The claim of a 'Material Turn' reflects an increased attention towards the object world, placing the 'thing-oriented aspect in the foreground of the description of culture and society'. Through the concept New Materialism, a new understanding of the independent being of the 'vibrant matter'[42] is carried forward, along with a critique of previous epistemology as being too engulfed in human subjectivism.[43] From a Design History perspective, we assert, obviously, that there is potential in focusing on objects. We also assert that the object world never disappeared from phenomenological or epistemological accounts of subjects-being-in-a-world-of-objects.[44] In this instance, the 'new in New Materialism seems to be largely a rhetorical gesture. For instance, Graham Harman's idea of an 'object-oriented ontology' opposing the withdrawal of all objects from relation'[45] seems to be a fight against a non-existent ghost. More problematically, the insistence on the material in New Materialism seems to be driven by a nostalgic wish to 'undo modernity' in a re-enchantment of the material world.[46]

Nevertheless, the material focus has led to interesting questions of *agencies and effects of materialities*. In this vein, Actor-Network Theory has emphasised the role and agency of non-human actors,[47] whether being attributed to a designed intention or not.

Generally speaking, the claim is that things *do* something on their own. We may take water drainage panels as an example. Being a part of modern cityscapes, they are designed to lead water away from the hard surfaces of cities and have in this sense been attributed with an agency of being transmission points between the surface of pavements and the underground sewers. A part of this agency is to absorb as much water as possible while filtering out large objects which could clog the system. In doing so, the panels are silent servants of water drainage in the cityscape. As a non-human actor in the network of water flows, water panels perform this task whether we acknowledge or notice them or not; we need not think of them and let them be an object of consciousness in order for them to be 'there' on the pavement.

By taking a closer look at specific water drainage panels, though, we can question the effects and performance of their material presence. In the case of *Orbit*, an ornamentally structured water drainage panel integrated in the pavements close to the stream in the Danish city of Aarhus (Fig. 5), the panel not only fulfils a function of leading water away – its aesthetics also attract attention by contributing to the flow, structure and identity of the actual space. When the human subject is always part of the equation, which is a tenet in aesthetic theory, aesthetic experience involves *someone* experiencing *something* that may be designed to elicit some kind of appreciation. In aesthetic terms we can scrutinize what is made present for sensory experience through the material being of *Orbit* and its actual material of cast iron and how this facilitates an encounter with (the meeting of) the world.

In its ornamented appearance, where layers of the panel are visible, *Orbit* offers a material structure for experiencing water. Aesthetically, it facilitates an encounter with water as a part of the cityscape by attracting attention to the process of letting it flow away underneath the pavement: it enables us to experience water as a visible part of the city. Seen through the lens of grey ecology, a panel like *Orbit* creates a stage for letting the 'cultural' artifice of the panel and the 'natural' element of water meet and exchange meanings: Materially, the panel is affected by letting water through (e.g. by rusting); conversely, the presence and being of water becomes a part of the aesthetic appearance of the city. Conceptually, the panel culturalises water as part of the space of the city. Hereby, the presence and potentiality of water (e.g. to overflow us) makes it clear that our cultural space of artificiality is always interacting with and dependent on natural elements, even if it often attempts to suppress or control them.

When grey ecology, in the words of Paul Virilio, is '*revelationary*'[48] in effect, it can, in the context of water and design, be understood as ways of getting dynamic interchanges of the fluid and the solid, the natural and the cultural, the uncontrollable and aim for control, out in the open and be acknowledged as part of the world as it is. In the same vein, Virilio coins the concept of 'trans-appearance'[49]: Just as materials, such as glass, can be non-opaque and transparent for what they transmit or contain, similarly, appearances can transcend themselves

('trans') and reveal their nature as part of an ongoing negotiation about what constitutes this world. On this basis, the materiality question for a design history of water would not only imply an analysis of changing material constitutions of designed objects. It would also imply an analysis of the ontological implication of the exchange of water and employed materials and, consequently, the epistemological question of how this exchange facilitates modes of gaining knowledge about water and design.

Temporal Flow

Since Heraclitus (lived c. 500 BC), water has been affiliated with – and been a metaphor for – change, time and human ageing, ritually staged in many practices, such as the Christian tradition of baptising childing by using water. On a profane-practical level, the design question in this instance concerns how designed solutions exercise and mediate cultural notions of the temporality of water. In this context, the question is how water, through design, can be mediated to represent different kinds of temporal modalities, ranging from the linear temporalities of flow to the circularity of closed circuits, making water appear eternally the same.

To begin with, virtually every flow of water, in the words of Paul Virilio regarding grey ecology, is an instance of 'contracting the distances of time'[50]: In principle, all water comes from the same global reservoir of sea and fresh water as part of a constant exchange eliminating the sequence of before, now and after. Water is always the same, albeit with different locations and presences. The aesthetic focus on design, then, aims at analysing how designed interfaces for handling water relate to this aspect of grey ecology and represent and/or exercise *various* temporal modalities of the flow of water. Ontologically as well as historically, we can contest the means of interaction at play in designed solutions for handling water while exploring how these solutions stage the experience of the temporal flow of water.

In the basic operation of cityscape fountains, water is presented in a constant flow which, paradoxically, at the same time eliminates the linearity of flow by creating a circular movement of water. In contrast, many modern drinking fountains, such as the 1,200 fountains installed since 2010 in Paris by the water supplier Eau de Paris,[51] require interactive gestures in order to work, thus staging the temporal movement of water while creating the need for designed interfaces for handling the fountain. Prior to handling fountains by pressing buttons or turning handles, drinking fountains could also let the water run, paradigmatically expressed in the so-called *nasoni* applied in Rome from the 1870s (and still being one of the signums of the city today) (Fig. 6). Through the nose of the fountain (hence the name), the water is constantly running at a slow pace. In some models there is also a hole at the top, so if the nose tap is blocked, the water springs from the hole to allow people to drink directly from the fountain.

On this point, we can differentiate between two kinds of experiences of the temporal flow of water. First, a primarily *visual* engagement can be fostered by objects staging a spectacular flow of water aimed at visual perception and appreciation. For instance, bathroom sinks with the tap on the one side and the drain on the other allow the water to flow across the sink, dramatising the effect of the swirling Coriolis force.

Second, *tactile* engagement with the flow of water may be facilitated in various ways. Water taps offer a sensual interaction and engagement with water: When you open the tap, the water flows to be used for some purpose. Interestingly, the development of hands-free sensor taps has changed tactile interaction to be with the water only, not the object. As an example, the Dyson Airblade Wash & Dry water tap (introduced 2019, Fig. 7) is an (almost) all-in-one hands-free object designed to facilitate the process of washing, automatically switching between water and air.[52] In this way, this object offers a tightly structured script for interaction, where the flow of water is replaced by brushes of air. Furthermore, the operation of the tap is highly dependent on the correct positioning of the hands. Thus, the tap, in spite of its rather simplified appearance, significantly *in*-scripts and disciplines its user.

In addition, the appearance of the tap, also being highly abstract in relation to understanding its purpose, could encourage a conceptual reflection of the sensual appearance in relating to its meaning. It is worth noting that the Dyson Airblade Wash and Dry water tap (or what we can call its object category) does not inscribe itself in a tradition of experimental or critical design, even if it incites a reflection about its being and purpose. Instead, its partial alienation – only partial due to its connection with a sink, indicating its purpose – inscribes it in a high-tech culture of ever-new, multifunctional and effortless objects. These objects are signums of innovation and a Modernist, reformist ideology of design as a promising means for improving human life. Basically, it is a product that celebrates technology. On the top of this, the brand value of 'Dyson' resonates with the product; its high-intensity air dryer demonstrates its kinship with other Dyson products rather than with other water taps. To nurture its brand value, the company has, as an example, sponsored design bloggers to travel to Dubai(!) and test the product and write a positive review,[53] inscribing the object in a culture of 'design' and not just as 'ordinary', helpful objects of everyday life, which, one could argue, most taps are.

While Dyson is living out the Modernist dream of efficient and effortless design in the service of creating better hygiene and cleanliness, many water-related solutions and practices have efforts inscribed in them as part of interacting with water. The mere fact that many people around the world do not have direct access to drinking water and spend a substantial amount of time just obtaining it is important to acknowledge when dealing with the question of effort and effortlessness. In this regard, an interesting category of design objects to examine would be pumps and pumping systems, which need to be investigated in their

political, social, cultural and economic context of (in)equalities in terms of accessing and distributing water.

Zooming in on one last example, the LifeStraw product family can be seen as an attempt to access drinking water in conditions and places without running water or with contaminated water. Originally developed by the end of the 1990s to filter out disease-evoking bacteria from water, especially the Guinea worm larva,[54] the LifeStraw has been designed to also filter out microplastics from water and to be used as outdoor gear for affluent people. Regardless of its entrance into this kind of lifestyle setting, which does not exclude its ability to improve life quality more widely, LifeStraw presents an interesting design solution due to its close, tactile interaction with water.

In its original design as a straw, the LifeStraw offers a direct tactile and gustatory interaction with both the product and the water. The user must place a high degree of trust in the product's ability to filter the water properly. In its operation, the product exemplifies the ontological negotiations at play in grey ecology when muddy water meets the interface of the 'grey' ordinariness of the straw. Ultimately, the product's ability to filter water can as a negotiation between life and death. Similarly, the Dyson product is also designed for hygienic purposes. By not having to touch the panel, the user may avoid catching disease or at least, post-COVID, think so when not touching something in a public space. At the same time, however, the Dyson product creates and enhances the distance between the designed interface and the use of water. In this sense, the product is not only alienating in appearance but also in relation to interacting with water. According to the dichotomy introduced by James Linton, the Dyson product can be seen to 'fixate' water, presenting it as an effortless and superficially aestheticised effect of the hyper-efficient product, whereas the LifeStraw encourages interaction with water 'in process'[55], fostering new ways of engaging with water based on its temporal flow.

Historicising Design for Water

Summing up, these conceptualisations of aesthetic modalities of sensory engagement, materiality and temporal flow provide a foundation for further investigations into the history of design for water. At the end of the day, the question of design for water is thus not just about changing objects of design related to various social and cultural contexts and locations. It is embedded in an epistemological reflection on how we, as humans, relate to our surroundings and, through the 'power tool' of design, create an artificial interface facilitating our being in the world. This interface may blur the lines between 'cultural' and 'natural' means. On one hand, water becomes a mirror of the culturalising effects of design; on the other hand, designed objects reveal the greyness of the water, its vital complexity as a medium for all life (and death) through their aesthetic modalities. Ultimately, the

culture/nature oppositions may prove to be shaky: we are nature ourselves, otherness exists within us, and culture and nature are never simply separable from each other.

In this instance, water is particularly interesting, as it is a substance we are most familiar with (and *the* most central element of our bodies). It extends the sphere of human cognition by being related to the unknowns of nature and of the more-than-human. Water existed long before humans, it is everywhere (even in the air), it exists on other planets, and it is a container for the life and history of all nature.

The aesthetic approach to design provides a phenomenological theory of sensing and understanding design-as-it-appears-to-us, feeding into an epistemological reflection of how we know the world through design. Again, designed objects and solutions are not the only access points to water, even if they can be difficult to avoid in highly artificial, urbanised environments. Further, the unitary human 'we' needs to be deconstructed and differentiated: In the perspective of a general epistemology, a human 'we' may be deployed as a unitary abstraction, but in actual social structures, 'we' are never the 'same' due to inequalities in terms of wealth and access to resources and variations in preference. Ideally, a future design history of water should encompass all these complexities and fractures.

Figure legends

Fig. 1. The Ice hotel in Jukkasjärvi (Sweden) demonstrates the use of water in the most solid of its three states (ice, water, steam). Photo: Asaf Kliger. Reproduced with permission.

Fig. 2. *Bankers Hygrometer*. Design by Arne Jacobsen (1970/71) in relation to his architectural work of the National Bank of Denmark. Reproduced with permission.

Fig. 3. Drinking glasses with design by Aino Aalto (1932) and Anders Raad (2021). Photo: Kirstine Autzen. Reproduced with permission.

Fig. 4. Commercial aestheticization of water in the example of the Norwegian company VOSS. Photo: @george_and_troja. Reproduced with permission.

Fig. 5. Ornamented water drainage panel *Orbit* close to the stream in the Danish city Aarhus. Design by Kristine Jensen. Manufactured since 2013 by GH Form. Reproduced with permission.

Fig. 6. Water posts in Rome implemented since the 1870s and called *nasone* (plural: *nasoni*), meaning nose, due to the shape and position of the tap. Photo: Wilfredo Hernandez. Reproduced due to Creative Commons CC0 1.0 Universal Public Domain Dedication.

Fig. 7. Dyson *Airblade Wash & Dry (WD04)* tap solution, introduced 2019. Image credit: Dyson. Reproduced with permission.

Endnotes

[1] Gernot Böhme and Hartmut Böhme, *Feuer, Wasser, Erde, Luft: Eine Kulturgeschichte der Elemente* (München: C.H. Beck, 1994).

[2] This is reflected in Immanuel Kant's guiding example of experiencing the sublime in nature when meeting 'the limitless ocean, set in uproar, [or] a high waterfall of a powerful river', *Kritik der Urtheilskraft* (Köln: Könemann, 1995), 130/§28.

[3] Jamie Linton, *What Is Water? The History of a Modern Abstraction* (Vancouver: UBC Press, 2010); see also David Manolo Sailer, *Die Disziplinierung des Wassers: Eine kultur- und ideengeschichtliche Analyse*. (Frankfurt a.M.: Peter Lang Verlag, 2022).

[4] See for instance, Ellen Lupton and J. Abbot Miller's *The Bathroom, the kitchen and the aesthetics of waste: A process of elimination*, published with Princeton Architectural Press and List Visual Arts center in 1992, or Barbara Penner's *Bathroom*, at Chicago University Press from 2013. More experimental and interdisciplinary works on this topic can be found in the 2024 second issue of the digital journal METODE, themed *Bathing, Bath and Beyond*: https://metode.r-o-m.no/en/volume/2.

[5] In contrast, the general cultural history of water has gained attention; cf. for instance, Hartmut Böhme, ed., *Kulturgeschichte des Wassers* (Frankfurt a.M.: Suhrkamp, 1988); Rila Mukherjee and Radhika Seshan, eds., 2015, "Special Issue: Approaching a History of Water: Tools and the Historian," *Water History* 7 (2), 2015; Dieter Gerten, *Wasser: Knappheit, Klimawandel, Welternährung* (München: C.H. Beck, 2020); Cecilia Chen, Janine MacLeod and Astrida Neimanis, eds., *Thinking with Water* (Montreal and Kingston: McGill-Queen's University Press, 2013); Steve Mentz, *Introduction to Blue Humanities*. (London: Routledge, 2024).

[6] The term was introduced by Donna Haraway in 2003 to critique ontological boundaries between species, gender, and living and non-living things. Since then, it has been developed and joined by a host of disparate expressions of what essentially are processual questions under the umbrella posthumanism. Haraway, Donna. *The Companion Species Manifesto: Dogs, people, and significant otherness* (Chicago: Prickly Paradigm Press, 2003). For a general overview of the term see Nicholas Malone and Kathryn Ovenden, "Natureculture," in *The International Encyclopedia of Primatology*, accessed 29 September 2024. DOI: 10.1002/9781119179313.wbprim0135

[7] Linton, *What is Water?*

[8] Ibid., 8.

[9] Ibid., 18.

[10] Ibid., 19.
[11] Ibid., 5.
[12] Cf., e.g., Penny Sparke, *An Introduction to Design and Culture: 1900 to the Present* (New York: Routledge. 2013).
[13] Claudia Mareis, *Theorien des Designs zur Einführung* (Hamburg: Junius, 2014).
[14] Herbert A. Simon, *The Sciences of the Artificial* (Cambridge, MA: MIT Press, 1996), 2.
[15] Otl Aicher, *die welt als entwurf* (Berlin: Ernst und Sohn, 1991), 196.
[16] Ibid., 185
[17] Clive Dilnot, "History, Design, Futures: Contending with What We Have Made," in *Design and the Question of History*, ed. Tony Fry, Clive Dilnot and Susan C. Steward (London: Bloomsbury, 2015), 174. Paradoxically, recent theories dealing with the "planetary" as a scale for design claim to acknowledge nature but end by positing design as a super-tool of human intention which, by consequence, risks neglecting the agency of nature (or the planet) as such, see, e.g., Ida Engholm, *Design for the New World* (London: Intellect, 2023). To a critical discussion, see Carsten Friberg, "Delusions about the human in the Anthropocene," *Nordicum-Mediterraneum* 19(2), 2024.
[18] Cf., e.g., Christina Cogdell, *Toward a living architecture? Complexism and biology in generative design* (Minneapolis: University of Minnesota Press, 2018).
[19] Gaston Bachelard, *L'Eau et les Rêves: Essai sur l'imagination de la matière* (Paris: José Corti, 1942).
[20] Ibid., 24.
[21] Ibid., 12.
[22] Cf. Jeremy Aynsley, *Designing Modern Germany* (London: Reaktion Books, 2009)
[23] Cf. Anthony Dunne, *Hertzian Tales: Electronic Products, Aesthetic Experiences, and Critical Design* (Cambridge, MA: MIT Press, 2005).
[24] Paul Virilio, *Grey Ecology* (New York & Dresden: Atropos, 2009), 71.
[25] Jeffrey Jerome Cohen, "Grey," in *Prismatic Ecologies*, ed. Jeffrey Jerome Cohen (Minneapolis: University of Minnesota Press, 2013), 271.
[26] Ibid., 273.
[27] Peter Sloterdijk, *Wer noch kein Grau gedacht hat: Eine Farbenlehre* (Frankfurt a.M.: Suhrkamp, 2023), 260.
[28] Cf., e.g., David Pye, *The Nature and Aesthetics of Design* (London: Herbert Press, 1978).
[29] Cf. Jane Forsey, *The Aesthetics of Design* (Oxford: Oxford University Press, 2013); Daniel Martin Feige. *Design: Eine philosophische Analyse* (Frankfurt am Main: Suhrkamp, 2018).
[30] Martin Seel, *Ästhetik des Erscheinens* (München: Hanser, 2000), 184, his italics.
[31] Ibid.
[32] Linton, *What is Water?*, 18.
[33] Vilém Flusser, *Dinge und Undinge: Phänomenologische Skizzen* (München: Carl Hanser Verlag, 1993), 13.
[34] Maurice Merleau-Ponty, *L'Œil et l'Esprit* (Paris: Gallimard, 1964), 70-1.

[35] Wolfgang Fritz Haug, *Kritik der Warenästhetik: Gefolgt von Warenästhetik im High-Tech-Kapitalismus* (Frankfurt am Main: Suhrkamp, 2009).
[36] "About," VOSS, accessed 22 April 2024, https://vosswater.com/about.
[37] Cf. Andreas Reckwitz, *Die Gesellschaft der Singularitäten—Zum Strukturwandel der Moderne* (Frankfurt am Main: Suhrkamp, 2019).
[38] Wolfgang Welsch, *Grenzgänge der Ästhetik* (Stuttgart: Reclam, 1996), 11.
[39] Ibid., 54.
[40] Cf. Terry Leddy, *The Extraordinary in the Ordinary: The Aesthetics of Everyday Life*. (Peterborough: Broadview Press, 2012).
[41] Linton*, What is Water?*, 4.
[42] Cf. Jane Bennett, *Vibrant Matter: A Political Ecology of Things* (Durham, NC: Duke University Press, 2010).
[43] See, for instance, Karen Barad, "Posthumanist Performativity: Toward an Understanding of How Matter Comes to Matter," *Signs: Journal of Women in Culture and Society* 28 (3), 2003, and Graham Harman, *Art and Objects* (Cambridge: Polity Press, 2020).
[44] In this way, for example, Maurice Merleau-Ponty's work can be read as an oscillation between investigating "inner", human consciousness and the phenomena of the "outer" world. See *Phénonénologie de la perception* (Paris: Gallimard, 1945) and *Le visible et l'invisible* (Paris: Gallimard, 1964).
[45] Harman, *Art and Objects*, 20.
[46] See Benjamin Boysen and Jesper Lundsfryd Rasmussen, "The Material Turn and the Fantasy to Undo Modernity," *The Comparatist* 44, 2020.
[47] E.g., Michel Callon, "Some Elements of a Sociology of Translation," in *The Science Studies Reader*, ed. M. Biagioli (New York: Routledge, 1986), 67–83; Bruno Latour, *Reassembling the Social* (Oxford: Oxford University Press, 2005).
[48] Virilio, *Grey Ecology*, 43, his italics.
[49] Ibid., 73.
[50] Paul Virilio, *L'Université du désastre* (Paris: Édition Galilée, 2007), 53.
[51] Cf. Karoline Margrethe Egholm and Cecilie Dollerup Lund, "Eau de Paris: Et bæredygtigt initiativ" (BA thesis, University of Southern Denmark, 2021).
[52] "Airblade Dry+Wash," Dyson, accessed 17 September 2024, https://www.dyson.com/commercial/hand-dryers/airblade-wash-dry-short.
[53] "Review," Lovethatdesign, published 2019 and accessed 29 August 2024, https://www.lovethatdesign.com/article/in-review-dysons-airblade-washdry.
[54] "Our Story," Lifestraw, accessed 29 August 2024, https://lifestraw.com/pages/our-story.
[55] Linton, *What is Water?*, 4.